

SAT UZBEKISTAN

Test creator Xodiyev Lutfulloxon

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Which of the following expressions cannot be equal to 0 for some value of x ?

- A) $x^2 - 2$
- B) $x^2 + 1$
- C) $1 - x^2$
- D) $2 - x^2$

2

$$f(x) = mx + b$$

In the function above, m and b are constants.

If $\frac{f(5) - f(2)}{3} = 2$, what is the values of m ?

- A) 2
- B) 3
- C) 4
- D) 5

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The line passing through the points $(a, 3)$ and $(b, -2)$ is parallel to the graph of $y = \frac{1}{2}x - 10$.

What is the value of $a - b$?

- A) 5
- B) 7
- C) 8
- D) 10

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4

$$\begin{aligned} y &= mx - \frac{2}{5} \\ 2x + 3y &= 4 \end{aligned}$$

In the system of equations, a is a constant. If the system has no solution, what is the value of m ?

- A) $-\frac{2}{3}$
- B) $-\frac{3}{2}$
- C) $\frac{2}{3}$
- D) $\frac{3}{2}$

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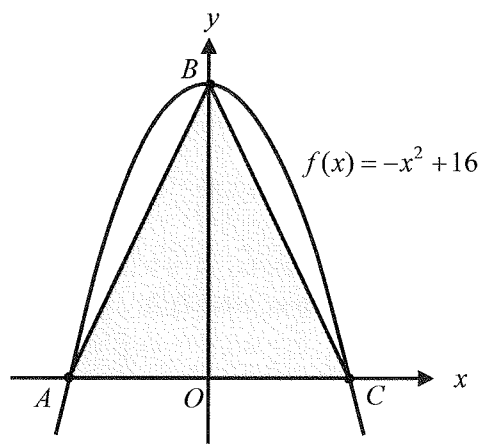
5

If $f(x) = (x-1)^2 - (x-1) - 1$, which of the following expressions is equal to $f(1-x)$?

- A) $f(1-x) = (x+1)^2 - (x+1) - 1$
- B) $f(1-x) = (1-x)^2 - (1-x) - 1$
- C) $f(1-x) = x^2 - x - 1$
- D) $f(1-x) = x^2 + x - 1$

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The graph of a function $f(x)$ is shown in the xy -plane above. What is the area of triangle ABC ?

- A) 16
- B) 32
- C) 64
- D) 128

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If $\frac{a-2b}{b} = \frac{2}{3}$, which of the following is equal to $\frac{a}{b}$?

- A) $\frac{2}{3}$
- B) $\frac{4}{3}$
- C) $\frac{5}{3}$
- D) $\frac{8}{3}$

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$$y = m\sqrt{x}$$

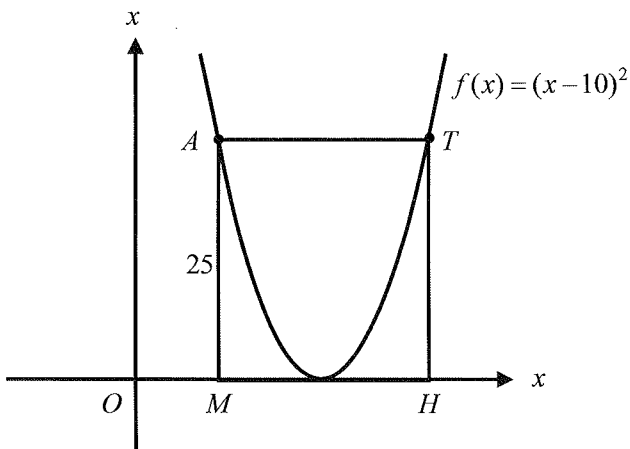
$$y = mx - k$$

In the system of equations, m and k are constants. If $(4, 16)$ is a solution to the system of equations above, what is the value of k ?

- A) 4
- B) 8
- C) 12
- D) 16

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The graphs of f and rectangle $MATH$ are shown in the xy -plane above. If $MA = 25$, what is the length of $\overline{AT}$?

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- A) 5
- B) 10
- C) 15
- D) 20

10

Which of the following equations has no solution?

- A) $10x - 5x = 3$
- B) $7x = 9x - 2x + 10$
- C) $10x - 6 = 8x + 2x - 6$
- D) $10x - 6 = 5x + 4x - 1$

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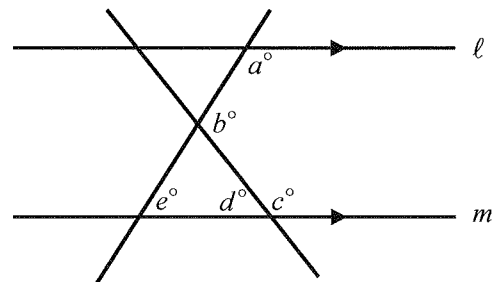
$$f(x) = k(x+4)(x-10)$$

In the quadratic function f above, k is a constant. The graph of the function in the xy -plane is a parabola with vertex (a, b) . If $b = -7$, which of the following is equal to k ?

- A) $\frac{1}{49}$
- B) $\frac{1}{7}$
- C) 7
- D) 49

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Note: Figure not drawn to scale.

In the figure above, if line ℓ and m are parallel, which of the following must be true?

- A) $a + b + c = 180$
- B) $d + e = b$
- C) $b + e = c$
- D) $a + b = 180$

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$$\frac{5x^2 + kx + 1}{x - 1} = ax + 1 + \frac{2}{x - 1}$$

The equation above is true for all values of x except 1, where k and a are constants. What is the value of k ?

- A) -4
- B) -2
- C) 2
- D) 4

14

What are the solutions to $4(x - 2)^2 - 1 = 5$?

- A) $x = 2 \pm \frac{\sqrt{6}}{4}$
- B) $x = 2 \pm \frac{\sqrt{6}}{2}$
- C) $x = \sqrt{2} \pm \frac{\sqrt{6}}{2}$
- D) $x = 2 \pm \sqrt{6}$

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Grade	For	Against	Total
Junior	60		
Senior			
Total	130		300

A supervisor surveyed students in his school to see if they were for or against building a fast-food restaurant in the school. The incomplete table above shows the results of his survey. If 40% of juniors are against it, how many seniors are in the school?

- A) 100
- B) 120
- C) 170
- D) 200

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